Nikita Shevyakov
Software Project Portfolio

Fall 2026

Nikita Shevyakov
B.S.E. Electrical and Computer Engineering | Computer Science Minor
Princeton University | Class of 2029

Selected Work

1. **CTRFL Research**

   Computational research on sustainable fuels with Ammonia-Hydrogen combustion

2. **ChipMasters Poker**

   A real-time multiplayer poker website with custom lobbies and graphics

3. **Virtual Obituary**

   A digital memorial for sharing various media memories of loved ones.

4. **DrawBattle**

   Multiplayer mobile game where two players battle in real time with colors to reach the opposite side first

github.com/Nikitaoverlord

linkedin.com/in/nikita-shevyakov • ns1962@princeton.edu

CTRFL Research

PRINCETON ENGINEERING

May – Aug 2026

Computational research on sustainable fuels with Ammonia-Hydrogen combustion

Python • Fortran • NumPy • pandas • Matplotlib

OVERVIEW

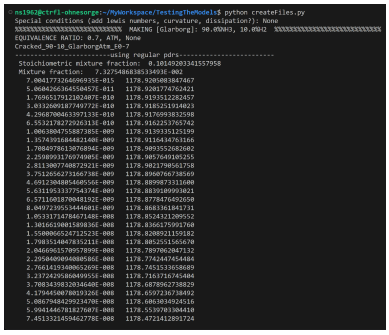
Using ammonia-hydrogen fuels for combustion is an established idea often discussed at the annual Combustion Conference. My job was to determine the ratio of ammonia, equivalence ratio, and temperature conditions to produce the least NO_x and N_2O, and then compare my simulation results with 2 other models. Additionally, I was to optimize and debug the existing codebase.

	Fuel Ratio	NH3															
	Condition	Atm						Turb						Atm			
	Equi Ratio	E0.7		E1.0		E1.4		E0.7		E1.0		E1.4		E0.7		E1.0	
	Spec	Norm	HiDis	Norm	HiDis	Norm	HiDis	Norm	HiDis	Norm	HiDis	Norm	HiDis	Norm	HiDis	Norm	HiDis
Molecule	Metric																
Y-OH	Peak Got	1.987e-03	**3.650e-04**	3.126e-03	**1.107e-03**	4.478e-05	**8.940e-05**	1.872e-03	**1.538e-03**	2.328e-03	**1.685e-03**	9.780e-05	**1.542e-04**	3.173e-03	**5.494e-04**	4.390e-03	**1.682e-03**
	Peak Glar	2.575e-03	**3.655e-04**	3.835e-03	**1.106e-03**	6.850e-05	**8.727e-05**	2.412e-03	**2.067e-03**	3.125e-03	**2.369e-03**	1.727e-04	**2.058e-04**	3.772e-03	**5.541e-04**	5.145e-03	**1.672e-03**
	Diff	-5.884e-04	**-4.573e-07**	-7.094e-04	**9.147e-07**	-2.372e-05	**2.135e-06**	-5.398e-04	**-5.292e-04**	-7.970e-04	**-6.839e-04**	-7.488e-05	**-5.156e-05**	-5.992e-04	**-4.710e-06**	-7.549e-04	**9.967e-06**
	%Diff	-2.580e+01	**-1.252e-01**	-2.038e+01	**8.265e-02**	-4.187e+01	**2.417e+00**	-2.520e+01	**-2.935e+01**	-2.923e+01	**-3.374e+01**	-5.537e+01	**-2.864e+01**	-1.726e+01	**-8.536e-01**	-1.583e+01	**5.944e-01**
Y-NO2	Peak Got	8.135e-06	**6.195e-06**	5.540e-06	**1.947e-06**	4.253e-06	**5.008e-07**	6.886e-05	**7.170e-05**	6.214e-05	**4.798e-05**	5.195e-05	**3.530e-05**	1.077e-05	**5.288e-06**	6.592e-06	**2.212e-06**
	Peak Glar	2.065e-05	**7.467e-06**	2.494e-05	**1.915e-06**	2.457e-05	**3.312e-06**	1.128e-04	**1.816e-04**	7.684e-05	**1.519e-04**	5.544e-05	**1.307e-04**	6.863e-05	**1.294e-05**	7.006e-05	**8.226e-06**
	Diff	-1.251e-05	**-1.271e-06**	-1.940e-05	**3.203e-08**	-2.031e-05	**-2.811e-06**	-4.399e-05	**-1.099e-04**	-1.470e-05	**-1.039e-04**	-3.490e-06	**-9.545e-05**	-5.786e-05	**-7.652e-06**	-6.347e-05	**-6.014e-06**
	%Diff	-8.695e+01	**-1.861e+01**	-1.273e+02	**1.659e+00**	-1.410e+02	**-1.475e+02**	-4.842e+01	**-8.677e+01**	-2.115e+01	**-1.040e+02**	-6.500e+00	**-1.150e+02**	-1.457e+02	**-8.396e+01**	-1.656e+02	**-1.152e+02**
Y-NO	Peak Got	3.833e-03	**1.786e-03**	4.446e-03	**8.911e-04**	3.183e-04	**1.576e-04**	4.509e-03	**4.509e-03**	4.086e-03	**2.984e-03**	6.189e-04	**6.957e-04**	6.057e-03	**2.244e-03**	6.491e-03	**1.395e-03**
	Peak Glar	5.875e-03	**1.787e-03**	6.061e-03	**9.053e-04**	5.925e-04	**1.951e-04**	7.346e-03	**6.949e-03**	7.025e-03	**5.947e-03**	1.697e-03	**1.744e-03**	8.405e-03	**2.238e-03**	8.181e-03	**1.530e-03**
	Diff	-2.042e-03	**-1.259e-06**	-1.615e-03	**-1.426e-05**	-2.742e-04	**-3.747e-05**	-2.837e-03	**-2.440e-03**	-2.940e-03	**-2.963e-03**	-1.078e-03	**-1.048e-03**	-2.347e-03	**5.754e-06**	-1.690e-03	**-1.349e-04**
	%Diff	-4.206e+01	**-7.050e-02**	-3.075e+01	**-1.588e+00**	-6.022e+01	**-2.124e+01**	-4.785e+01	**-4.258e+01**	-5.291e+01	**-6.636e+01**	-9.312e+01	**-8.592e+01**	-3.246e+01	**2.568e-01**	-2.304e+01	**-9.225e+00**
Y-N2O	Peak Got	3.789e-03	**7.023e-05**	2.347e-03	**1.212e-04**	6.739e-04	**1.577e-05**	3.819e-03	**5.272e-03**	1.872e-03	**1.778e-03**	1.219e-03	**5.713e-04**	3.944e-03	**2.623e-04**	2.960e-03	**3.252e-04**
	Peak Glar	3.272e-03	**2.537e-05**	2.299e-03	**5.940e-05**	1.055e-03	**1.589e-05**	4.778e-03	**5.188e-03**	3.496e-03	**2.684e-03**	3.012e-03	**1.212e-03**	3.785e-03	**1.563e-04**	3.052e-03	**2.183e-04**
	Diff	5.167e-04	**4.486e-05**	4.808e-05	**6.182e-05**	-3.809e-04	**-1.151e-07**	-9.589e-04	**8.493e-05**	-1.624e-03	**-9.058e-04**	-1.793e-03	**-6.404e-04**	1.589e-04	**1.061e-04**	-9.234e-05	**1.069e-04**
	%Diff	1.463e+01	**9.384e+01**	2.070e+00	**6.846e+01**	-4.407e+01	**-7.270e-01**	-2.231e+01	**1.624e+00**	-6.050e+01	**-4.060e+01**	-8.476e+01	**-7.184e+01**	4.112e+00	**5.069e+01**	-3.072e+00	**3.934e+01**

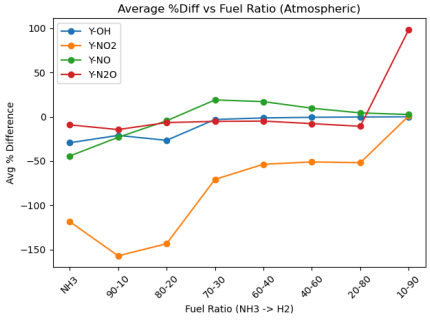
HIGHLIGHTS

- Debugged 20,000+ line Fortran codebase to support new combustion simulator version
- Wrote a Python script to automate the input file creation and simulation, turning hours of work into a single line terminal command.
- Used pandas and matplotlib to create presentable data tables and charts for final report

Running the Python automatic input-data generator for all fuel ratios



Running the Python automated input-data program



[Github](#) • [Report](#)

ChipMasters Poker

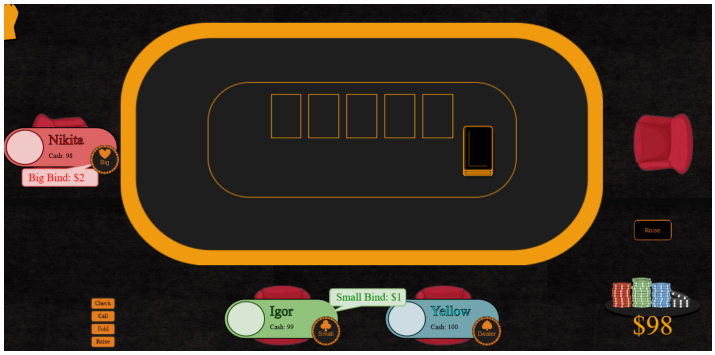
2025 – 2026

A real-time multiplayer poker website with custom lobbies and graphics

JavaScript • Express • Socket.IO • HTML/CSS • JSON Data Storage

OVERVIEW

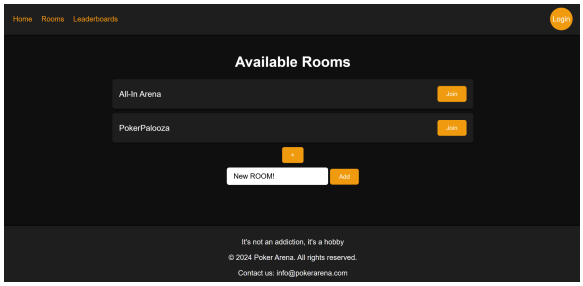
Create a poker account and join with your friends connected on the same WiFi network: create your own poker lobbies, customize your name tags, and play until everyone runs out of cash (non legal tender)! Originally deployed on chipmasters.sneakyzone.com with Akamai.



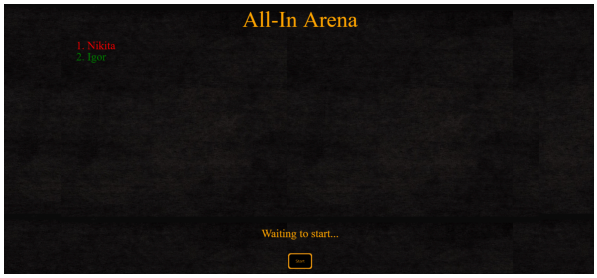
HIGHLIGHTS

- Real-time updates between players using Socket.IO and data storage for user sessions at later times
- Ability to create new rooms, player accounts, and full poker game progression and logic
- Fully self-made graphical UI and animations

Pick or create new game rooms and servers



See new players in the lobby before starting game



[Github](#) • [View Demo](#)

Virtual Obituary

2024 – 2025

A virtual memoriam with easily sharable digital content of passed loved ones

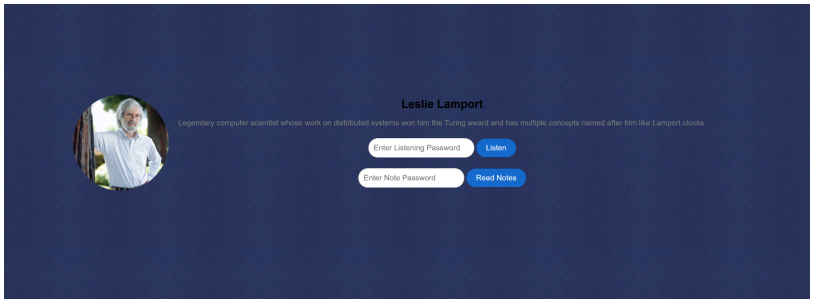
Javascript • Express • Socket.IO • HTML/CSS

Why I Built it

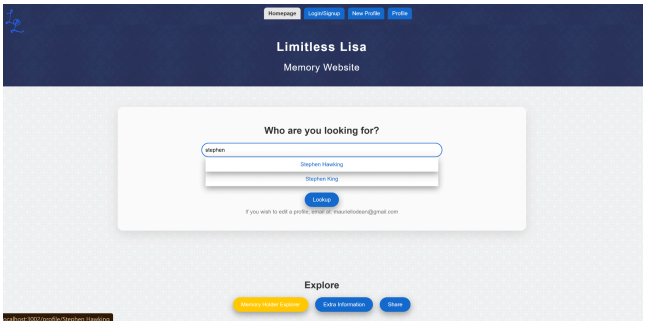
My coding partner and I developed Virtual Obituary for our Design Projects class after he had recently lost his mother to ALS. We wanted to go beyond traditional paper obituaries with a lasting digital memorial where family and friends could share photos, videos, audio, and personal notes.

What We Built

We designed a muti-page website with Express that generated new subdomains for each new profile. I focused on the backend creating a forum for adding images, videos, notes to profiles, as well as implementing account creation. I also created a search engine to filter for specific people and a scrollable feed for new additions. My partner focused on frontend design, including the sites' graphics and information page.







[Github]

DrawBattle

2022 – 2023

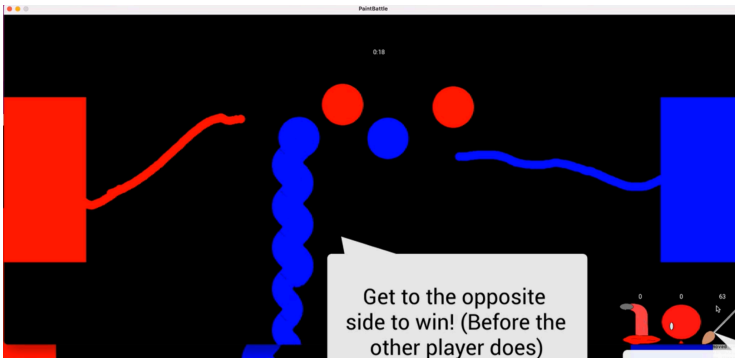
Multiplayer mobile game where two players battle in real time with colors to reach the opposite side first
Python • Kivy • Socket • Pickle

OVERVIEW

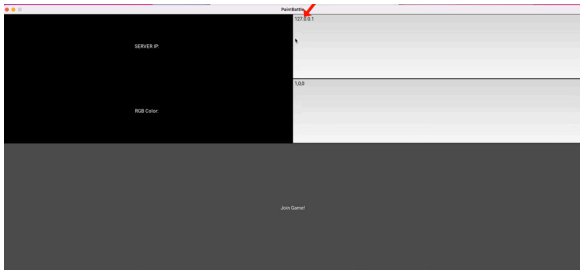
A cross-device multiplayer game made using the Python Kivy library and sockets. The game involves two opposing colored players with different “paint tools” trying to color their way to the opposing side first.

HIGHLIGHTS

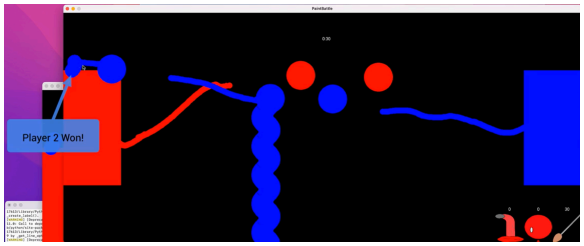
- Connect to a common server with its IP address if on the same WiFi network using Python sockets
- Play against 1 other user in real time (draw, shoot balloons, waterfall) with server-side game logic and client-side key and mouse input
- Can be distributed on mobile devices due to the GUI being coded with the Kivy library
- All hand drawn images and independently made game concepts



Join a game by connecting to the server IP and customize your color



Paint your way to the opposite color by throwing paint balloons, “paintfalls”, or drawing



[Github](#) • [Demo](#)